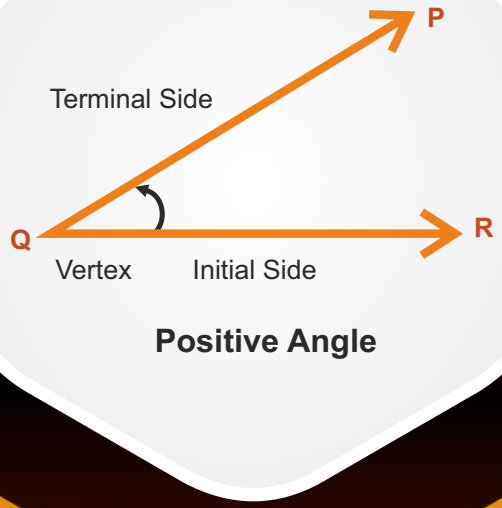
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Compound Angles

(English Medium)

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TRIGONOMETRIC RATIOS & IDENTITIES

1. INTRODUCTION TO TRIGONOMETRY :

The word 'trigonometry' is derived from the Greek words 'trigon' and 'metron' and it means 'measuring the sides of a triangle'. The subject was originally developed to solve geometric problems involving triangles. It was studied by sea captains for navigation, surveyor to map out the new lands, by engineers and others. Currently, trigonometry is used in many areas such as the science of seismology, designing electric circuits, describing the state of an atom, predicting the heights of tides in the ocean, analysing a musical tone and in many other areas.

(a) **Measurement of angles :** Commonly two systems of measurement of angles are used.

(i) **Sexagesimal or English System :** Here 1 right angle = 90° (degrees)

$1° = 60'$ (minutes)

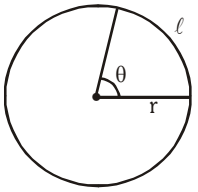
$1' = 60''$ (seconds)

(ii) **Circular system :** Here an angle is measured in radians. One radian corresponds to the angle subtended by an arc of length 'r' at the centre of the circle of radius r. It is a constant quantity and does not depend upon the radius of the circle.

(b) Relation between the these systems : $\frac{D}{90} = \frac{R}{\pi/2}$

(c) If θ is the angle subtended at the centre of a circle of radius 'r',

by an arc of length 'ℓ' then $\frac{\ell}{r} = \theta$.



Note that here ℓ, r are in the same units and θ is always in radians.

Illustration 1 : If the arcs of same length in two circles subtend angles of 60° and 75° at their centres. Find the ratio of their radii.

Solution : Let r_1 and r_2 be the radii of the given circles and let their arcs of same length 's' subtend angles of 60° and 75° at their centres.

$$\text{Now, } 60° = \left(60 \times \frac{\pi}{180}\right)^c = \left(\frac{\pi}{3}\right)^c \text{ and } 75° = \left(75 \times \frac{\pi}{180}\right)^c = \left(\frac{5\pi}{12}\right)^c$$

$$\therefore \quad \frac{\pi}{3} = \frac{s}{r_1} \text{ and } \frac{5\pi}{12} = \frac{s}{r_2}$$

$$\Rightarrow \quad \frac{\pi}{3} r_1 = s \text{ and } \frac{5\pi}{12} r_2 = s \Rightarrow \frac{\pi}{3} r_1 = \frac{5\pi}{12} r_2 \Rightarrow 4r_1 = 5r_2 \Rightarrow r_1 : r_2 = 5 : 4 \quad \textbf{Ans.}$$

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Do yourself - 1 :

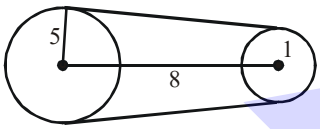
1. The radius of a circle is 30 cm. Find the length of smaller arc of the circle if the length of the chord of the arc is 30 cm.

2. A man is running around a regular hexagonal field of side length 6m, so that he always at a distance 3 metre from the nearest boundary. Find the length of path travelled by him in one round.

3. Convert the following measurement into radians :

(a) 25º 30' 30" (b) 10º 42' 30" (c) 9º 18' 42"

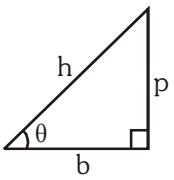
4. A belt is tied up across two circular pulleys of radii 5m and 1m respectively whose centres are seperated at a distance 8m. (as shown). Find the length of the belt required.



5. Find the number of degrees, minutes and seconds in the angle at the centre of a circle, whose radius is 5m, which is subtended by an arc of length 6 m. (Consider $\pi = 22/7$)

2. T-RATIOS (or Trigonometric functions) :

In a right angle triangle



$\sin\theta = \frac{p}{h}$; $\cos\theta = \frac{b}{h}$; $\tan\theta = \frac{p}{b}$; $\operatorname{cosec}\theta = \frac{h}{p}$; $\sec\theta = \frac{h}{b}$ and $\cot\theta = \frac{b}{p}$

'p' is perpendicular ; 'b' is base and 'h' is hypotenuse.

3. BASIC TRIGONOMETRIC IDENTITIES :

(1) $\sin\theta.\operatorname{cosec}\theta = 1$ **(2)** $\cos\theta.\sec\theta = 1$

(3) $\tan\theta.\cot\theta = 1$ **(4)** $\tan\theta = \frac{\sin\theta}{\cos\theta}$ & $\cot\theta = \frac{\cos\theta}{\sin\theta}$

(5) $\sin^2\theta + \cos^2\theta = 1$ or $\sin^2\theta = 1 - \cos^2\theta$ or $\cos^2\theta = 1 - \sin^2\theta$

(6) $\sec^2\theta - \tan^2\theta = 1$ or $\sec^2\theta = 1 + \tan^2\theta$ or $\tan^2\theta = \sec^2\theta - 1$

(7) $\sec\theta + \tan\theta = \frac{1}{\sec\theta - \tan\theta}$

(8) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$ or $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$ or $\cot^2\theta = \operatorname{cosec}^2\theta - 1$

(9) $\operatorname{cosec}\theta + \cot\theta = \frac{1}{\operatorname{cosec}\theta - \cot\theta}$

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(10) Expressing trigonometrical ratio in terms of each other :

	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\cot\theta$	$\sec\theta$	$\operatorname{cosec}\theta$
$\sin\theta$	$\sin\theta$	$\sqrt{1-\cos^2\theta}$	$\frac{\tan\theta}{\sqrt{1+\tan^2\theta}}$	$\frac{1}{\sqrt{1+\cot^2\theta}}$	$\frac{\sqrt{\sec^2\theta-1}}{\sec\theta}$	$\frac{1}{\operatorname{cosec}\theta}$
$\cos\theta$	$\sqrt{1-\sin^2\theta}$	$\cos\theta$	$\frac{1}{\sqrt{1+\tan^2\theta}}$	$\frac{\cot\theta}{\sqrt{1+\cot^2\theta}}$	$\frac{1}{\sec\theta}$	$\frac{\sqrt{\operatorname{cosec}^2\theta-1}}{\operatorname{cosec}\theta}$
$\tan\theta$	$\frac{\sin\theta}{\sqrt{1-\sin^2\theta}}$	$\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}$	$\tan\theta$	$\frac{1}{\cot\theta}$	$\sqrt{\sec^2\theta-1}$	$\frac{1}{\sqrt{\operatorname{cosec}^2\theta-1}}$
$\cot\theta$	$\frac{\sqrt{1-\sin^2\theta}}{\sin\theta}$	$\frac{\cos\theta}{\sqrt{1-\cos^2\theta}}$	$\frac{1}{\tan\theta}$	$\cot\theta$	$\frac{1}{\sqrt{\sec^2\theta-1}}$	$\sqrt{\operatorname{cosec}^2\theta-1}$
$\sec\theta$	$\frac{1}{\sqrt{1-\sin^2\theta}}$	$\frac{1}{\cos\theta}$	$\sqrt{1+\tan^2\theta}$	$\frac{\sqrt{1+\cot^2\theta}}{\cot\theta}$	$\sec\theta$	$\frac{\operatorname{cosec}\theta}{\sqrt{\operatorname{cosec}^2\theta-1}}$
$\operatorname{cosec}\theta$	$\frac{1}{\sin\theta}$	$\frac{1}{\sqrt{1-\cos^2\theta}}$	$\frac{\sqrt{1+\tan^2\theta}}{\tan\theta}$	$\sqrt{1+\cot^2\theta}$	$\frac{\sec\theta}{\sqrt{\sec^2\theta-1}}$	$\operatorname{cosec}\theta$

Illustration 2 : If $\sin\theta+\sin^2\theta=1$, then prove that $\cos^{12}\theta+3\cos^{10}\theta+3\cos^8\theta+\cos^6\theta-1=0$

Solution : Given that $\sin\theta = 1 - \sin^2\theta = \cos^2\theta$

L.H.S. $= \cos^6\theta(\cos^2\theta + 1)^3 - 1 = \sin^3\theta(1 + \sin\theta)^3 - 1 = (\sin\theta + \sin^2\theta)^3 - 1 = 1 - 1 = 0$

Illustration 3 : $4(\sin^6\theta + \cos^6\theta) - 6(\sin^4\theta + \cos^4\theta)$ is equal to

(A) 0 (B) 1 (C) –2 (D) none of these

Solution : $4[(\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)] - 6[(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta]$

$= 4[1 - 3\sin^2\theta\cos^2\theta] - 6[1 - 2\sin^2\theta\cos^2\theta]$

$= 4 - 12\sin^2\theta\cos^2\theta - 6 + 12\sin^2\theta\cos^2\theta = -2$ **Ans.(C)**

Do yourself - 2 :

1. If $\cot\theta = \frac{4}{3}$, then find the value of $\sin\theta$, $\cos\theta$ and $\operatorname{cosec}\theta$ in first quadrant.

2. If $\sin\theta + \operatorname{cosec}\theta = 2$, then find the value of $\sin^8\theta + \operatorname{cosec}^8\theta$

Prove the following statements in their respective valid domains :

3. $\cos^4A - \sin^4A + 1 = 2\cos^2A$

4. $(\sin A + \cos A)(1 - \sin A\cos A) = \sin^3A + \cos^3A$.

5. $\frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} = 2\operatorname{cosec} A$

6. $\cos^6A + \sin^6A = 1 - 3\sin^2A\cos^2A$.

7. $\sqrt{\frac{1-\sin A}{1+\sin A}} = \sec A - \tan A \quad (-90^\circ < A < 90^\circ)$

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8. $\dfrac{1}{\cot A+\tan A}=\sin A\cos A$

9. $\dfrac{1-\tan A}{1+\tan A}=\dfrac{\cot A-1}{\cot A+1}$

10. $\dfrac{\tan A}{1-\cot A}+\dfrac{\cot A}{1-\tan A}=\sec A\,\text{cosec}\,A+1$

11. $\dfrac{\cos A}{1-\tan A}+\dfrac{\sin A}{1-\cot A}=\sin A+\cos A.$

12. $\sec^4A-\sec^2A=\tan^4A+\tan^2A.$

13. $\cot^4A+\cot^2A=\text{cosec}^4A-\text{cosec}^2A.$

14. $\tan^2A-\sin^2A=\sin^4A\sec^2A=\tan^2A.\sin^2A$

15. $(1+\cot A-\text{cosec}A)(1+\tan A+\sec A)=2$

16. $\dfrac{1}{\text{cosec}\,A-\cot A}-\dfrac{1}{\sin A}=\dfrac{1}{\sin A}-\dfrac{1}{\text{cosec}\,A+\cot A}$

17. $\dfrac{\cot A.\cos A}{\cot A+\cos A}=\dfrac{\cot A-\cos A}{\cot A.\cos A}$

18. $\left(\dfrac{1}{\sec^2\alpha-\cos^2\alpha}+\dfrac{1}{\text{cosec}^2\alpha-\sin^2\alpha}\right)\cos^2\alpha\sin^2\alpha=\dfrac{1-\cos^2\alpha\sin^2\alpha}{2+\cos^2\alpha\sin^2\alpha}$

19. $\sin^8A-\cos^8A=(\sin^2A-\cos^2A)(1-2\sin^2A\cos^2A)$

20. $\dfrac{\tan A+\sec A-1}{\tan A-\sec A+1}=\dfrac{1+\sin A}{\cos A}$

21. $(\tan\alpha+\text{cosec}\,\beta)^2-(\cot\beta-\sec\alpha)^2=2\tan\alpha\cot\beta(\text{cosec}\,\alpha+\sec\beta).$

22. $2\sec^2\alpha-\sec^4\alpha-2\,\text{cosec}^2\alpha+\text{cosec}^4\alpha=\cot^4\alpha-\tan^4\alpha.$

23. If $\sin\theta$ equals to $\dfrac{x^2-y^2}{x^2+y^2}$, find the value of $\cos\theta$ and $\cot\theta$, where $\theta\in(0,90^\circ)$.

24. If $\sin\theta=\dfrac{m^2+2mn}{m^2+2mn+2n^2}$, prove that $\tan\theta=\dfrac{m^2+2mn}{2mn+2n^2}$, where $\theta\in(0,90^\circ)$.

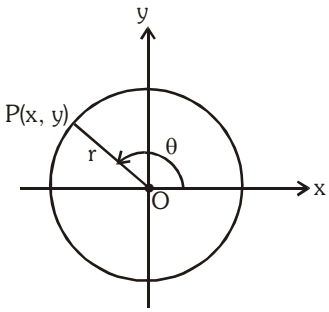
25. If $\cos\theta-\sin\theta=\sqrt{2}\sin\theta$, prove that $\cos\theta+\sin\theta=\sqrt{2}\cos\theta$.

26. Prove that $\text{cosec}^6\alpha-\cot^6\alpha=3\,\text{cosec}^2\alpha\cot^2\alpha+1.$

27. Express $2\sec^2A-\sec^4A-2\text{cosec}^2A+\text{cosec}^4A$ in terms of $\tan A$.

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4. NEW DEFINITION OF T-RATIOS :



By using rectangular coordinates the definitions of trigonometric functions can be extended to angles of any size in the following way (see diagram). A point P is taken with coordinates (x, y). The radius vector OP has length r and the angle θ is taken as the directed angle measured anticlockwise from the x-axis. The three main trigonometric functions are then defined in terms of r and the coordinates x and y.

$\sin\theta = y/r$,

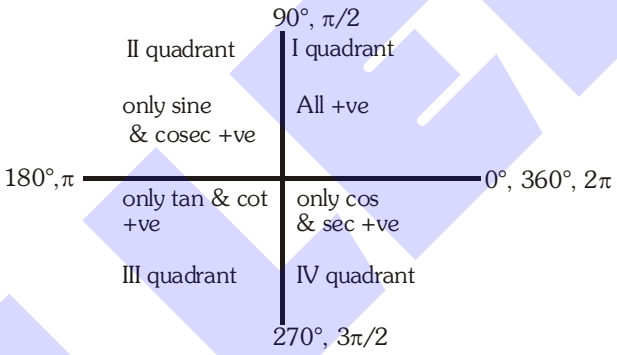
$\cos\theta = x/r$

$\tan\theta = y/x$,

(The other function are reciprocals of these)

This can give negative values of the trigonometric functions.

5. SIGNS OF TRIGONOMETRIC FUNCTIONS IN DIFFERENT QUADRANTS :



6. TRIGONOMETRIC FUNCTIONS OF ALLIED ANGLES :

(a) $\sin(2n\pi + \theta) = \sin\theta$, $\cos(2n\pi + \theta) = \cos\theta$, where $n \in I$

(b)

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$
$\sin(90° - \theta) = \cos\theta$	$\cos(90° - \theta) = \sin\theta$
$\sin(90° + \theta) = \cos\theta$	$\cos(90° + \theta) = -\sin\theta$
$\sin(180° - \theta) = \sin\theta$	$\cos(180° - \theta) = -\cos\theta$
$\sin(180° + \theta) = -\sin\theta$	$\cos(180° + \theta) = -\cos\theta$
$\sin(270° - \theta) = -\cos\theta$	$\cos(270° - \theta) = -\sin\theta$
$\sin(270° + \theta) = -\cos\theta$	$\cos(270° + \theta) = \sin\theta$
$\sin(360° - \theta) = -\sin\theta$	$\cos(360° - \theta) = \cos\theta$
$\sin(360° + \theta) = \sin\theta$	$\cos(360° + \theta) = \cos\theta$

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Do yourself - 3 :

1. If $\sin A = \frac{11}{61}$, find tanA, cosA, and sec A.

2. If $\cos\theta = \frac{4}{5}$, find sin θ and cot θ.

3. If $\tan\theta = \frac{1}{\sqrt{7}}$, find the value of $\frac{\text{cosec}^2\theta - \sec^2\theta}{\text{cosec}^2\theta + \sec^2\theta}$

4. If $\cot\theta = \frac{15}{8}$, find sin θ and cosec θ.

5. If 2 sin θ = 2 – cos θ, find sin θ.

6. If 8 sin θ = 4 + cos θ, find sin θ.

7. If tan θ + sec θ = 1.5, find sin θ.

8. If cot θ + cosec θ = 5, find cos θ.

9. If $3\sec^4\theta + 8 = 10\sec^2\theta$, find the values of tan θ.

10. If $\tan^2\theta + \sec\theta = 5$, find cos θ.

11. If tan θ + cot θ = 2, find sin θ.

12. If $\sec^2\theta = 2 + 2\tan\theta$, find tan θ.

13. If $\tan\theta = \frac{2x(x+1)}{2x+1}$, find sin θ and cos θ.

7. VALUES OF T-RATIOS OF SOME STANDARD ANGLES :

Angles	**0°**	**30°**	**45°**	**60°**	**90°**	**180°**	**270°**
T-ratio	**0**	**π/6**	**π/4**	**π/3**	**π/2**	**π**	**3π/2**
sin θ	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0	–1
cos θ	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	–1	0
tan θ	0	$1/\sqrt{3}$	1	$\sqrt{3}$	N.D.	0	N.D.
cot θ	N.D.	$\sqrt{3}$	1	$1/\sqrt{3}$	0	N.D.	0
sec θ	1	$2/\sqrt{3}$	$\sqrt{2}$	2	N.D.	–1	N.D.
cosecθ	N.D.	2	$\sqrt{2}$	$2/\sqrt{3}$	1	N.D.	–1

N.D. → Not Defined

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$ where $n \in I$

(b) $\sin(2n+1)\frac{\pi}{2} = (-1)^n$; $\cos(2n+1)\frac{\pi}{2} = 0$ where $n \in I$

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Do yourself - 4 :

1. Verify the following identities for A = 30º as well as for A = 45º

(a) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

(b) $\sin 2A = 2\sin A \cos A$ (c) $\cos 3A = 4\cos^3 A - 3\cos A$

(d) $\sin 3A = 3\sin A - 4\sin^3 A$ (e) $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

2. Find the value of

(a) $\sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ$ (b) $\tan^2 30^\circ + \tan^2 45^\circ + \tan^2 60^\circ$

(c) $\sin 30^\circ \cos 60^\circ + \sin 30^\circ \sin 60^\circ$ (d) $\cos 45^\circ \cos 60^\circ - \sin 45^\circ \sin 60^\circ$

Illustration 4 : If $\sin\theta = -\dfrac{1}{2}$ and $\tan\theta = \dfrac{1}{\sqrt{3}}$ then θ is equal to -

(A) 30° (B) 150° (C) 210° (D) none of these

Solution : Let us first find out θ lying between 0 and 360°.

Since $\sin\theta = -\dfrac{1}{2} \Rightarrow \theta = 210^\circ$ or 330° and $\tan\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$ or 210°

Hence , $\theta = 210^\circ$ or $\dfrac{7\pi}{6}$ is the value satisfying both. **Ans. (C)**

Do yourself - 5 :

1. (i) If $\cos\theta = -\dfrac{1}{2}$ and $\pi < \theta < \dfrac{3\pi}{2}$, then find the value of $4\tan^2\theta - 3\operatorname{cosec}^2\theta$.

(ii) Prove that : (a) $\cos 570^\circ \sin 510^\circ + \sin(-330^\circ)\cos(-390^\circ) = 0$

(b) $\tan\dfrac{11\pi}{3} - 2\sin\dfrac{9\pi}{3} - \dfrac{3}{4}\operatorname{cosec}^2\dfrac{\pi}{4} + 4\cos^2\dfrac{17\pi}{6} = \dfrac{3-2\sqrt{3}}{2}$

2. Evaluate :

(a) $\sin 420^\circ \cos 390^\circ + \cos(-300^\circ)\sin(-330^\circ)$ (b) $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ$

3. What are the values of cosA – sinA and tanA + cotA when A has the values

(a) $\dfrac{2\pi}{3}$ (b) $\dfrac{7\pi}{4}$ (c) $\dfrac{11\pi}{3}$ (d) $\dfrac{5\pi}{4}$

4. Express the following quantities in terms of the ratios of a positive angle, which is less than 45º

(a) sin(–65º) (b) cos(–928º) (c) tan1145º (d) cot(–1054º)

5. What is the sign of sinA + cosA for the following values of A ?

(a) 140º (b) –356º (c) –1125º.

6. What is the sign of sinA – cosA for the following values of A ?

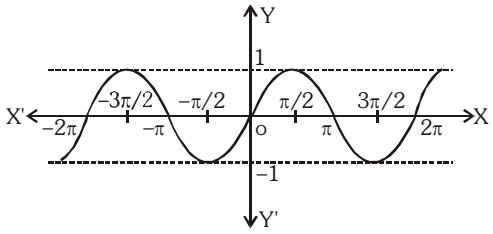
(a) 215º (b) –634º (c) – 457º

7. Find the sines and cosines of all angles in the first four quadrant whose tangents are equal to cos 135º.

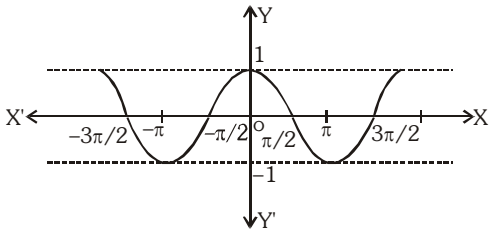
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8. GRAPH OF TRIGONOMETRIC FUNCTIONS :

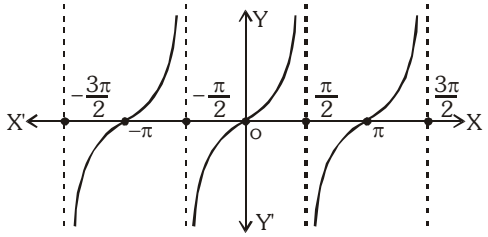
(i) y = sinx



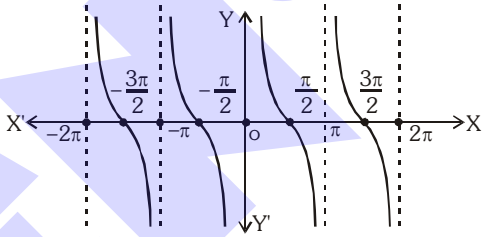
(ii) y = cosx



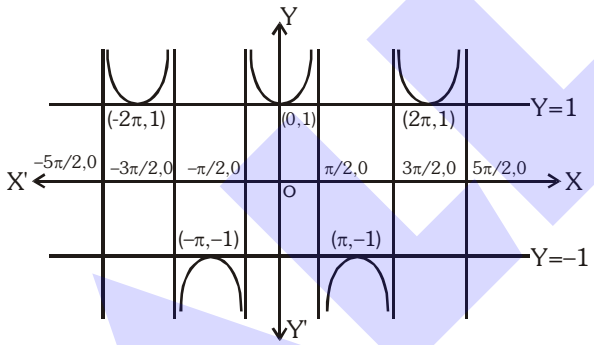
(iii) y = tanx



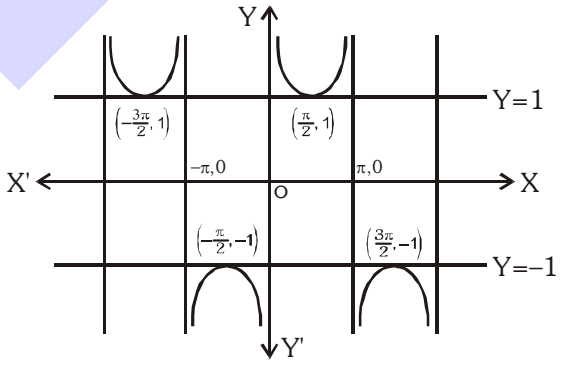
(iv) y = cotx



(v) y = secx



(vi) y = cosecx



9. DOMAINS, RANGES AND PERIODICITY OF TRIGONOMETRIC FUNCTIONS :

T-Ratio	Domain	Range	Period
sin x	R	[–1,1]	2π
cos x	R	[–1,1]	2π
tan x	R–{(2n+1)π/2 ; n∈I}	R	π
cot x	R–{nπ : n ∈ I}	R	π
sec x	R– {(2n+1) π/2 : n ∈ I}	(–∞,–1] ∪[1,∞)	2π
cosec x	R– {nπ : n ∈ I}	(–∞,–1] ∪[1,∞)	2π

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Do yourself - 6 :

1. Prove that the equation $\sin\theta = x + \frac{1}{x}$ is impossible if x be real.

2. Show that the equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is only possible when x = y.

3. The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is -
(A) 1 (B) 0 (C) 2 (D) Infinite

4. Draw graphs of
(a) $y = \sin 2x$ (b) $y = 2\cos 3x$ (c) $y = 4\tan x$

5. Find number of solutions of the equation $\sin(\pi x) + x^2 + 1 = 2x$

10. TRIGONOMETRIC RATIOS OF THE SUM & DIFFERENCE OF TWO ANGLES :

(i) $\sin(A + B) = \sin A\cos B + \cos A\sin B$. **(ii)** $\sin(A - B) = \sin A\cos B - \cos A\sin B$.

(iii) $\cos(A + B) = \cos A\cos B - \sin A\sin B$ **(iv)** $\cos(A - B) = \cos A\cos B + \sin A\sin B$

(v) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$ **(vi)** $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$

(vii) $\cot(A + B) = \frac{\cot B\cot A - 1}{\cot B + \cot A}$ **(viii)** $\cot(A - B) = \frac{\cot B\cot A + 1}{\cot B - \cot A}$

Some more results :

(i) $\sin^2 A - \sin^2 B = \sin(A + B).\sin(A - B) = \cos^2 B - \cos^2 A$.

(ii) $\cos^2 A - \sin^2 B = \cos(A+B).\cos(A - B)$.

Illustration 5 : Prove that $\sqrt{3}\,\text{cosec}20° - \sec 20° = 4$.

Solution : $\text{L.H.S.} = \frac{\sqrt{3}}{\sin 20°} - \frac{1}{\cos 20°} = \frac{\sqrt{3}\cos 20° - \sin 20°}{\sin 20°.\cos 20°}$

$$= \frac{4\left(\frac{\sqrt{3}}{2}\cos 20° - \frac{1}{2}\sin 20°\right)}{2\sin 20°\cos 20°} = \frac{4(\sin 60.\cos 20° - \cos 60°.\sin 20°)}{\sin 40°}$$

$$= 4.\frac{\sin(60° - 20°)}{\sin 40°} = 4.\frac{\sin 40°}{\sin 40°} = 4 = \text{R.H.S.}$$

Illustration 6 : Prove that $\tan 70° = \cot 70° + 2\cot 40°$.

Solution : $\text{L.H.S.} = \tan 70° = \tan(20° + 50°) = \frac{\tan 20° + \tan 50°}{1 - \tan 20°\tan 50°}$

$\text{or } \tan 70° - \tan 20°\tan 50°\tan 70° = \tan 20° + \tan 50°$

$\text{or } \tan 70° = \tan 70°\tan 50°\tan 20° + \tan 20° + \tan 50° = 2\tan 50° + \tan 20°$

$= \cot 70° + 2\cot 40° \quad = \text{R.H.S.}$

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Do yourself - 7 :

1. If $\sin A = \frac{3}{5}$ and $\cos B = \frac{9}{41}$, $0 < A, B < \frac{\pi}{2}$, then find the value of the following :

(a) sin(A + B) (b) sin(A – B) (c) cos(A + B) (d) cos(A – B)

2. If x + y = 45°, then prove that :

(a) (1 + tanx)(1 + tany) = 2 (b) (cotx – 1)(coty – 1) = 2

(Remember these results)

3. If $\sin\alpha = \frac{3}{5}$ and $\cos\beta = \frac{9}{41}$, find the value of sin(α – β) and cos(α + β).

4. If $\sin\alpha = \frac{15}{17}$ and $\cos\beta = \frac{12}{13}$, find the values of sin(α + β), cos(α – β), and tan(α + β)

Prove that

5. cos(45° – A) cos(45° – B) – sin(45° – A) sin(45° – B) = sin(A + B).

6. sin(45° + A) cos(45° – B) + cos(45° + A) sin(45° – B) = cos(A – B).

7. $\frac{\sin(A-B)}{\cos A\cos B} + \frac{\sin(B-C)}{\cos B\cos C} + \frac{\sin(C-A)}{\cos C\cos A} = 0$

8. sin105° + cos105° = cos45°

9. sin75° – sin15° = cos105° + cos15°

10. cos(α + β) cosγ – cos(β + γ)cosα = sinβ sin(γ – α)

11. sin(n + 1)A sin(n – 1) A + cos(n + 1)A cos(n – 1)A = cos2A.

12. sin(n + 1)A sin(n + 2) A + cos(n + 1)A cos(n + 2)A = cosA.

13. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, find the values of tan(2A + B) and tan(2A – B)

14. If $\tan\alpha = \frac{5}{6}$ and $\tan\beta = \frac{1}{11}$, prove that $\alpha + \beta = \frac{\pi}{4}$

Prove that

15. $\tan\left(\frac{\pi}{4}+\theta\right)\times\tan\left(\frac{3\pi}{4}+\theta\right) = -1$

16. $\cot\left(\frac{\pi}{4}+\theta\right)\cot\left(\frac{\pi}{4}-\theta\right) = 1$

17. $1 + \tan A\tan\frac{A}{2} = \tan A\cot\frac{A}{2} - 1 = \sec A$

18. tan3A tan2A tanA = tan3A – tan2A – tanA.

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11. FORMULAE TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE :

(i) $2\sin A\cos B = \sin(A+B)+\sin(A-B)$. **(ii)** $2\cos A\sin B = \sin(A+B)-\sin(A-B)$

(iii) $2\cos A\cos B = \cos(A+B)+\cos(A-B)$ **(iv)** $2\sin A\sin B = \cos(A-B)-\cos(A+B)$

Illustration 7 : If $\sin2A = \lambda\sin2B$, then prove that $\dfrac{\tan(A+B)}{\tan(A-B)} = \dfrac{\lambda+1}{\lambda-1}$.

Solution : Given $\sin2A = \lambda\sin2B$

$$\Rightarrow \quad \frac{\sin 2A}{\sin 2B} = \frac{\lambda}{1}$$

Applying componendo & dividendo,

$$\frac{\sin 2A+\sin 2B}{\sin 2B-\sin 2A} = \frac{\lambda+1}{1-\lambda}$$

$$\Rightarrow \quad \frac{2\sin\left(\frac{2A+2B}{2}\right)\cos\left(\frac{2A-2B}{2}\right)}{2\cos\left(\frac{2B+2A}{2}\right)\sin\left(\frac{2B-2A}{2}\right)} = \frac{\lambda+1}{1-\lambda}$$

$$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin\{-(A-B)\}} = \frac{\lambda+1}{1-\lambda} \quad \Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\times-\sin(A-B)} = \frac{\lambda+1}{-(\lambda-1)}$$

$$\Rightarrow \quad \frac{\sin(A+B)\cos(A-B)}{\cos(A+B)\sin(A-B)} = \frac{\lambda+1}{\lambda-1} \quad \Rightarrow \quad \tan(A+B)\cot(A-B) = \frac{\lambda+1}{\lambda-1}$$

$$\Rightarrow \quad \frac{\tan(A+B)}{\tan(A-B)} = \frac{\lambda+1}{\lambda-1}$$

Do yourself - 8 :

1. Simplify $\dfrac{\sin 75^\circ-\sin 15^\circ}{\cos 75^\circ+\cos 15^\circ}$

Express follwing as a sum or difference of angles used in arguments :

2. $2\sin5\theta\sin7\theta$.

3. $2\sin54^\circ\sin66^\circ$

4. $\cos(36^\circ-A)\cos(36^\circ+A)+\cos(54^\circ+A)\cos(54^\circ-A) = \cos2A$.

5. $\cos A\sin(B-C)+\cos B\sin(C-A)+\cos C\sin(A-B) = 0$.

6. $\sin(45^\circ+A)\sin(45^\circ-A) = \dfrac{1}{2}\cos2A$

7. $\sin(\beta-\gamma)\cos(\alpha-\delta)+\sin(\gamma-\alpha)\cos(\beta-\delta)+\sin(\alpha-\beta)\cos(\gamma-\delta) = 0$.

8. $2\cos\dfrac{\pi}{13}\cos\dfrac{9\pi}{13}+\cos\dfrac{3\pi}{13}+\cos\dfrac{5\pi}{13} = 0$

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12. FORMULAE TO TRANSFORM SUM OR DIFFERENCE INTO PRODUCT :

(i) $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(ii) $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(iii) $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(iv) $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$

Illustration 8 : $\frac{\sin 5\theta + \sin 2\theta - \sin\theta}{\cos 5\theta + 2\cos 3\theta + 2\cos^2\theta + \cos\theta}$ is equal to -

(A) $\tan\theta$ (B) $\cos\theta$ (C) $\cot\theta$ (D) none of these

Solution : L.H.S.$= \frac{2\sin 2\theta\cos 3\theta + \sin 2\theta}{2\cos 3\theta.\cos 2\theta + 2\cos 3\theta + 2\cos^2\theta} = \frac{\sin 2\theta[2\cos 3\theta + 1]}{2\left[\cos 3\theta(\cos 2\theta + 1) + (\cos^2\theta)\right]}$

$= \frac{\sin 2\theta[2\cos 3\theta + 1]}{2\left[\cos 3\theta(2\cos^2\theta) + \cos^2\theta\right]} = \frac{\sin 2\theta(2\cos 3\theta + 1)}{2\cos^2\theta(2\cos 3\theta + 1)} = \tan\theta$ **Ans. (A)**

Do yourself - 9 :

1. Prove that

(a) $(\sin 3A + \sin A)\sin A + (\cos 3A - \cos A)\cos A = 0$ **(b)** $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta} = \tan 2\theta$

Prove that

2. $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan\theta$

3. $\frac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A} = \cot(A+B)\cot(A-B)$

4. $\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \frac{\tan(A+B)}{\tan(A-B)}$

5. $\cos(A + B) + \sin(A - B) = 2\sin(45° + A)\cos(45° + B)$

6. $\frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \frac{\sin A}{\cos 2A\cos 3A}$

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7. $\dfrac{\tan 5\theta+\tan 3\theta}{\tan 5\theta-\tan 3\theta}=4\cos 2\theta\cos 4\theta$

8. $\dfrac{\cos 3\theta+2\cos 5\theta+\cos 7\theta}{\cos\theta+2\cos 3\theta+\cos 5\theta}=\cos 2\theta-\sin 2\theta\tan 3\theta$

9. $\dfrac{\sin A+\sin 3A+\sin 5A+\sin 7A}{\cos A+\cos 3A+\cos 5A+\cos 7A}=\tan 4A$

10. $\dfrac{\sin(A-C)+2\sin A+\sin(A+C)}{\sin(B-C)+2\sin B+\sin(B+C)}=\dfrac{\sin A}{\sin B}$

11. $\dfrac{\sin A-\sin 5A+\sin 9A-\sin 13A}{\cos A-\cos 5A-\cos 9A+\cos 13A}=\cot 4A$

12. $\dfrac{\sin A+\sin B}{\sin A-\sin B}=\tan\dfrac{A+B}{2}\cot\dfrac{A-B}{2}$

13. $\dfrac{\sin A+\sin B}{\cos A+\cos B}=\tan\dfrac{A+B}{2}$

14. $\dfrac{\sin A-\sin B}{\cos B-\cos A}=\cot\dfrac{A+B}{2}$

15. $\dfrac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)}=\cot B$

16. $\cos\left\{\theta+\left(n-\dfrac{3}{2}\right)\phi\right\}-\cos\left\{\theta+\left(n+\dfrac{3}{2}\right)\phi\right\}=2\sin\dfrac{3\phi}{2}.\sin(\theta+n\phi)$

17. $\sin\dfrac{\theta}{2}\sin\dfrac{7\theta}{2}+\sin\dfrac{3\theta}{2}\sin\dfrac{11\theta}{2}=\sin 2\theta\sin 5\theta$

18. $\cos 2\theta\cos\dfrac{\theta}{2}-\cos 3\theta\cos\dfrac{9\theta}{2}=\sin 5\theta\sin\dfrac{5\theta}{2}$

19. $\dfrac{2\sin(A-C)\cos C-\sin(A-2C)}{2\sin(B-C)\cos C-\sin(B-2C)}=\dfrac{\sin A}{\sin B}$

20. $\dfrac{\sin A\sin 2A+\sin 3A\sin 6A+\sin 4A\sin 13A}{\sin A\cos 2A+\sin 3A\cos 6A+\sin 4A\cos 13A}=\tan 9A$

21. $\dfrac{\cos 2A\cos 3A-\cos 2A\cos 7A+\cos A\cos 10A}{\sin 4A\sin 3A-\sin 2A\sin 5A+\sin 4A\sin 7A}=\cot 6A\cot 5A$

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13. TRIGONOMETRIC RATIOS OF SUM OF MORE THAN TWO ANGLES :

(i) $\sin(A+B+C) = \sin A\cos B\cos C + \sin B\cos A\cos C + \sin C\cos A\cos B - \sin A\sin B\sin C$

$= \Sigma \sin A \cos B \cos C - \Pi \sin A$

$= \cos A \cos B \cos C [\tan A + \tan B + \tan C - \tan A \tan B \tan C]$

(ii) $\cos(A+B+C) = \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C$

$= \Pi \cos A - \Sigma \sin A \sin B \cos C$

$= \cos A \cos B \cos C [1 - \tan A \tan B - \tan B \tan C - \tan C \tan A]$

(iii) $\tan(A + B + C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = \dfrac{S_1 - S_3}{1 - S_2}$

14. TRIGONOMETRIC RATIOS OF MULTIPLE ANGLES :

(a) Trigonometrical ratios of an angle 2θ in terms of the angle θ :

(i) $\sin 2\theta = 2 \sin\theta \cos\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

(ii) $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

(iii) $1 + \cos 2\theta = 2\cos^2\theta$ **(iv)** $1 - \cos 2\theta = 2\sin^2\theta$

(v) $\tan\theta = \dfrac{1-\cos 2\theta}{\sin 2\theta} = \dfrac{\sin 2\theta}{1+\cos 2\theta}$ **(vi)** $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$

Illustration 9: Prove that : $\dfrac{2\cos 2A + 1}{2\cos 2A - 1} = \tan(60° + A)\tan(60° - A)$.

Solution : R.H.S. $= \tan(60° + A)\tan(60° - A)$

$$= \left(\frac{\tan 60° + \tan A}{1 - \tan 60° \tan A}\right)\left(\frac{\tan 60° - \tan A}{1 + \tan 60° \tan A}\right) = \left(\frac{\sqrt{3} + \tan A}{1 - \sqrt{3}\tan A}\right)\left(\frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}\right)$$

$$= \frac{3 - \tan^2 A}{1 - 3\tan^2 A} = \frac{3 - \dfrac{\sin^2 A}{\cos^2 A}}{1 - 3\dfrac{\sin^2 A}{\cos^2 A}} = \frac{3\cos^2 A - \sin^2 A}{\cos^2 A - 3\sin^2 A} == \frac{2\cos^2 A + \cos^2 A - 2\sin^2 A + \sin^2 A}{2\cos^2 A - 2\sin^2 A - \sin^2 A - \cos^2 A}$$

$$= \frac{2(\cos^2 A - \sin^2 A) + \cos^2 A + \sin^2 A}{2(\cos^2 A - \sin^2 A) - (\sin^2 A + \cos^2 A)} = \frac{2\cos 2A + 1}{2\cos 2A - 1} = \text{L.H.S.}$$

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Do yourself - 10 :

Prove that :

1. $\frac{\sin 2A}{1+\cos 2A} = \tan A$ (Remember)

2. $\frac{\sin 2A}{1-\cos 2A} = \cot A$ (Remember)

3. tanA – cotA = –2 cot2A. (Remember)

4. $\frac{1+\sin 2\theta+\cos 2\theta}{1+\sin 2\theta-\cos 2\theta} = \cot\theta$

5. tanA + cotA = 2 cosec2A

6. $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)} = \tan\frac{A}{2}\cot\frac{B}{2}$

7. $\frac{\sec 8A-1}{\sec 4A-1} = \frac{\tan 8A}{\tan 2A}$

8. $\frac{1+\tan^2(45°-A)}{1-\tan^2(45°-A)} = \text{cosec } 2A$

9. $\frac{\sin^2 A-\sin^2 B}{\sin A\cos A-\sin B\cos B} = \tan(A+B)$

10. $\tan\left(\frac{\pi}{4}+\theta\right)-\tan\left(\frac{\pi}{4}-\theta\right) = 2\tan 2\theta$

11. $\cot(A + 15°) - \tan(A - 15°) = \frac{4\cos 2A}{1+2\sin 2A}$

12. $\frac{\sin(n+1)A+2\sin nA+\sin(n-1)A}{\cos(n-1)A-\cos(n+1)A} = \cot\frac{A}{2}$

13. $\sin3A + \sin2A - \sin A = 4 \sin A \cos \frac{A}{2}\cos\frac{3A}{2}$

14. $\tan 2A = (\sec 2A+1)\sqrt{\sec^2 A-1}$

15. $\cos^3 2\theta + 3\cos 2\theta = 4(\cos^6\theta - \sin^6\theta)$

16. $1+ \cos^2 2\theta = 2(\cos^4\theta + \sin^4\theta)$

17. $\sec^2 A(1 + \sec 2A) = 2 \sec 2A$

18. $\text{cosec}A - 2\cot 2A \cos A = 2\sin A$

19. $\cot A = \frac{1}{2}\left(\cot\frac{A}{2}-\tan\frac{A}{2}\right)$

20. $\frac{2\cos 2^n\theta+1}{2\cos\theta+1} = (2\cos\theta-1)(2\cos 2\theta-1)(2\cos 2^2\theta-1)....(2\cos 2^{n-1}\theta-1)$

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(b) Trigonometrical ratios of an angle 3θ in terms of the angle θ :

(i) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. **(ii)** $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

(iii) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

Illustration 10: Show that $\sin 12^\circ . \sin 48^\circ . \sin 54^\circ = 1/8$

Solution : $\text{L.H.S.} = \dfrac{1}{2}[\cos 36^\circ - \cos 60^\circ]\sin 54^\circ = \dfrac{1}{2}\left[\cos 36^\circ \sin 54^\circ - \dfrac{1}{2}\sin 54^\circ\right]$

$$= \frac{1}{4}[2\cos 36^\circ \sin 54^\circ - \sin 54^\circ] = \frac{1}{4}[\sin 90^\circ + \sin 18^\circ - \sin 54^\circ]$$

$$= \frac{1}{4}[1 - (\sin 54^\circ - \sin 18^\circ)] = \frac{1}{4}[1 - 2\sin 18^\circ \cos 36^\circ]$$

$$= \frac{1}{4}\left[1 - \frac{2\sin 18^\circ}{\cos 18^\circ}\cos 18^\circ \cos 36^\circ\right] = \frac{1}{4}\left[1 - \frac{\sin 36^\circ \cos 36^\circ}{\cos 18^\circ}\right]$$

$$= \frac{1}{4}\left[1 - \frac{2\sin 36^\circ \cos 36^\circ}{2\cos 18^\circ}\right] = \frac{1}{4}\left[1 - \frac{\sin 72^\circ}{2\sin 72^\circ}\right] = \frac{1}{4}\left[1 - \frac{1}{2}\right] = \frac{1}{8} = \text{R.H.S.}$$

Illustration 11: Prove that : $\tan A + \tan(60^\circ + A) + \tan(120^\circ + A) = 3\tan 3A$

Solution : $\text{L.H.S.} = \tan A + \tan(60^\circ + A) + \tan(120^\circ + A)$

$= \tan A + \tan(60^\circ + A) + \tan\{180^\circ - (60^\circ - A)\}$

$= \tan A + \tan(60^\circ + A) - \tan(60^\circ - A)$ $\quad [\because \tan(180^\circ - \theta) = -\tan\theta]$

$$= \tan A + \frac{\tan 60^\circ + \tan A}{1 - \tan 60^\circ \tan A} - \frac{\tan 60^\circ - \tan A}{1 + \tan 60^\circ \tan A} = \tan A + \frac{\sqrt{3} + \tan A}{1 - \sqrt{3}\tan A} - \frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}$$

$$= \tan A + \frac{\sqrt{3} + \tan A + 3\tan A + \sqrt{3}\tan^2 A - \sqrt{3} + \tan A + 3\tan A - \sqrt{3}\tan^2 A}{(1 - \sqrt{3}\tan A)(1 + \sqrt{3}\tan A)}$$

$$= \tan A + \frac{8\tan A}{1 - 3\tan^2 A} = \frac{\tan A - 3\tan^3 A + 8\tan A}{1 - 3\tan^2 A}$$

$$= \frac{9\tan A - 3\tan^3 A}{1 - 3\tan^2 A} = 3\left(\frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}\right) = 3\tan 3A = \text{R.H.S.}$$

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Do yourself - 11 :

1 Prove that :

(a) $\cot\theta \cot(60^\circ - \theta) \cot(60^\circ + \theta) = \cot 3\theta$ (b) $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

(c) $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

Prove that

2. $\sin\alpha \sin(60^\circ - \alpha) \sin(60^\circ + \alpha) = \frac{1}{4}\sin 3\alpha$

3. $\cos\alpha \cos(60^\circ - \alpha) \cos(60^\circ + \alpha) = \frac{1}{4}\cos 3\alpha$

4. $\cot\alpha + \cot(60^\circ + \alpha) - \cot(60^\circ - \alpha) = 3\cot 3\alpha$.

5. $\cos 4\alpha = 1 - 8\cos^2\alpha + 8\cos^4\alpha$

6. $\sin 4A = 4\sin A \cos^3 A - 4\cos A \sin^3 A$.

7. $\cos 6\alpha = 32\cos^6\alpha - 48\cos^4\alpha + 18\cos^2\alpha - 1$.

8. If $\cos x + \sin x = a, \left(-\frac{\pi}{2} < x < -\frac{\pi}{4}\right)$, then cos2x is equal to

(A) a^2 (B) $a\sqrt{(2-a)}$ (C) $a\sqrt{(2+a)}$ (D) $a\sqrt{(2-a^2)}$

9. If $\cos A = \frac{3}{4}$, then the value of expression $32\sin\frac{A}{2}\sin\frac{5A}{2}$ is equal to

(A) 11 (B) –11 (C) 12 (D) 4

15. TRIGONOMETRIC RATIOS OF SUB MULTIPLE ANGLES :

Since the trigonometric relations are true for all values of angle θ, they will be true if instead of θ be substitute $\frac{\theta}{2}$

(i) $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(ii) $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = 2\cos^2\frac{\theta}{2} - 1 = 1 - 2\sin^2\frac{\theta}{2} = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

(iii) $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ **(iv)** $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$

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(v) $\tan\frac{\theta}{2}=\frac{1-\cos\theta}{\sin\theta}=\frac{\sin\theta}{1+\cos\theta}$ **(vi)** $\tan\theta=\frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$

(vii) $\sin\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{2}}$ **(viii)** $\cos\frac{\theta}{2}=\pm\sqrt{\frac{1+\cos\theta}{2}}$

(ix) $\tan\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$ **(x)** $2\sin\frac{\theta}{2}=\pm\sqrt{1+\sin\theta}\pm\sqrt{1-\sin\theta}$

(xi) $2\cos\frac{\theta}{2}=\pm\sqrt{1+\sin\theta}\mp\sqrt{1-\sin\theta}$ **(xii)** $\tan\frac{\theta}{2}=\frac{\pm\sqrt{1+\tan^2\theta}-1}{\tan\theta}$

for **(vii)** to **(xii)** , we decide the sign of ratio according to value of θ.

Illustration 12: $\sin\ 67\frac{1}{2}^\circ+\cos 67\frac{1}{2}^\circ$ is equal to

(A) $\frac{1}{2}\sqrt{4+2\sqrt{2}}$ (B) $\frac{1}{2}\sqrt{4-2\sqrt{2}}$ (C) $\frac{1}{4}\left(\sqrt{4+2\sqrt{2}}\right)$ (D) $\frac{1}{4}\left(\sqrt{4-2\sqrt{2}}\right)$

Solution : $\sin 67\frac{1}{2}^\circ+\cos 67\frac{1}{2}^\circ=\sqrt{1+\sin 135^\circ}=\sqrt{1+\frac{1}{\sqrt{2}}}$ (using cosA + sinA = $\sqrt{1+\sin 2A}$)

$=\frac{1}{2}\sqrt{4+2\sqrt{2}}$ **Ans.(A)**

Do yourself - 12 :

1 Find the value of

(a) $\sin\frac{\pi}{8}$ (b) $\cos\frac{\pi}{8}$ (c) $\tan\frac{\pi}{8}$

2. $\frac{\cos A}{1\mp\sin A}=\tan\left(45^\circ\pm\frac{A}{2}\right)$

3. $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}=\tan\frac{\theta}{2}$

4. If $\sin\theta=\frac{1}{2}$ and $\sin\phi=\frac{1}{3}$, find the values of sin (θ + ϕ) and sin(2θ + 2ϕ).

5. If $\cos\alpha=\frac{11}{61}$ and $\sin\beta=\frac{4}{5}$, find the values of $\sin^2\frac{\alpha-\beta}{2}$ and $\cos^2\frac{\alpha+\beta}{2}$, the angles α and β being positive acute angles.

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6. If $\cos\alpha = \frac{3}{5}$ and $\cos\beta = \frac{4}{5}$, find the value of $\cos\frac{\alpha-\beta}{2}$, the angles α and β being positive acute angles.

7. Given $\sec\theta = 1\frac{1}{4}$, find $\tan\frac{\theta}{2}$ and $\tan\theta$.

8. Find the values of (a) $\sin 7\frac{1}{2}^\circ$ (b) $\cos 7\frac{1}{2}^\circ$ (c) $\tan 11\frac{1}{4}^\circ$

9. If $\sin\theta + \sin\phi = a$ and $\cos\theta + \cos\phi = b$, find the value of $\tan\frac{\theta-\phi}{2}$.

Prove that

10. $\left(\cos\alpha+\cos\beta\right)^2+\left(\sin\alpha-\sin\beta\right)^2 = 4\cos^2\frac{\alpha+\beta}{2}$.

11. $\left(\cos\alpha+\cos\beta\right)^2+\left(\sin\alpha+\sin\beta\right)^2 = 4\cos^2\frac{\alpha-\beta}{2}$.

12. $\left(\cos\alpha-\cos\beta\right)^2+\left(\sin\alpha-\sin\beta\right)^2 = 4\sin^2\frac{\alpha-\beta}{2}$

13. $\sec\left(\frac{\pi}{4}+\theta\right)\sec\left(\frac{\pi}{4}-\theta\right) = 2\sec 2\theta$

14. $\tan\left(45^\circ+\frac{A}{2}\right) = \sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$

15. $\sin^2\left(\frac{\pi}{8}+\frac{A}{2}\right)-\sin^2\left(\frac{\pi}{8}-\frac{A}{2}\right) = \frac{1}{\sqrt{2}}\sin A$

16. $\cos^2\alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ) = \frac{3}{2}$

17. $\cos^4\frac{\pi}{8}+\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}+\cos^4\frac{7\pi}{8} = \frac{3}{2}$

18. $\sin^4\frac{\pi}{8}+\sin^4\frac{3\pi}{8}+\sin^4\frac{5\pi}{8}+\sin^4\frac{7\pi}{8} = \frac{3}{2}$

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16. TRIGONOMETRIC RATIOS OF SOME STANDARD ANGLES :

(i) $\sin 18^\circ = \sin\frac{\pi}{10} = \frac{\sqrt{5}-1}{4} = \cos 72^\circ = \cos\frac{2\pi}{5}$

(ii) $\cos 36^\circ = \cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4} = \sin 54^\circ = \sin\frac{3\pi}{10}$

(iii) $\sin 72^\circ = \sin\frac{2\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18^\circ = \cos\frac{\pi}{10}$

(iv) $\sin 36^\circ = \sin\frac{\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ = \cos\frac{3\pi}{10}$

(v) $\sin 15^\circ = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ = \cos\frac{5\pi}{12}$

(vi) $\cos 15^\circ = \cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ = \sin\frac{5\pi}{12}$

(vii) $\tan 15^\circ = \tan\frac{\pi}{12} = 2-\sqrt{3} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \cot 75^\circ = \cot\frac{5\pi}{12}$

(viii) $\tan 75^\circ = \tan\frac{5\pi}{12} = 2+\sqrt{3} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \cot 15^\circ = \cot\frac{\pi}{12}$

(ix) $\tan(22.5^\circ) = \tan\frac{\pi}{8} = \sqrt{2}-1 = \cot(67.5^\circ) = \cot\frac{3\pi}{8}$

(x) $\tan(67.5^\circ) = \tan\frac{3\pi}{8} = \sqrt{2}+1 = \cot(22.5^\circ) = \cot\frac{\pi}{8}$

Illustration 13: Evaluate sin78° – sin66° – sin42° + sin6°.

Solution : The expression $= (\sin 78^\circ - \sin 42^\circ) - (\sin 66^\circ - \sin 6^\circ) = 2\cos(60^\circ)\sin(18^\circ) - 2\cos 36^\circ . \sin 30^\circ$

$= \sin 18^\circ - \cos 36^\circ = \left(\frac{\sqrt{5}-1}{4}\right) - \left(\frac{\sqrt{5}+1}{4}\right) = -\frac{1}{2}$

Do yourself - 13 :

1 Find the value of

(a) $\sin\frac{\pi}{10} + \sin\frac{13\pi}{10}$ (b) $\cos^2 48^\circ - \sin^2 12^\circ$

Evaluate :

2. $\sin^2 72^\circ - \sin^2 60^\circ$

3. $\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$

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4. $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ$

5. $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{3\pi}{15}\cos\frac{4\pi}{15}\cos\frac{5\pi}{15}\cos\frac{6\pi}{15}\cos\frac{7\pi}{15}$

6. $16\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{8\pi}{15}\cos\frac{14\pi}{15}$

7. Two parallel chords of a circle, which are on the same side of the centre, subtend angles of 72° and 144° respectively at the centre. Prove that the perpendicular distance between the chords is half the radius of the circle.

8. In any circle prove that the chord which subtends 108° at the centre is equal to the sum of the two chords which subtend angles of 36° and 60°.

9. If $A = \cos\frac{\pi}{9}\cos\frac{4\pi}{9}\cos\frac{16\pi}{9}, B = \cos\frac{2\pi}{9}\cos\frac{8\pi}{9}\cos\frac{32\pi}{9}$, then value of $\left|\frac{1}{16A.B}\right|$ is

10. The value of $\cos\frac{\pi}{10}\cos\frac{2\pi}{10}\cos\frac{4\pi}{10}\cos\frac{8\pi}{10}\cos\frac{16\pi}{10}$ is

(A) $\frac{\sqrt{10+2\sqrt{5}}}{64}$ (B) $-\frac{\cos(\pi/10)}{16}$ (C) $\frac{\cos(\pi/10)}{16}$ (D) $-\frac{\sqrt{10+2\sqrt{5}}}{16}$

17. CONDITIONAL TRIGONOMETRIC IDENTITIES :

If $A + B + C = 180^\circ$, then

(i) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(ii) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

(iii) $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

(iv) $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$

(v) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(vi) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(vii) $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

(viii) $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

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Illustration 14: In any triangle ABC, sin A – cos B = cos C, then angle B is (Where $\angle B > \angle C$)

(A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

Solution : We have , sin A – cos B = cos C

sin A = cos B + cos C

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2}=2\cos\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2}=2\cos\left(\frac{\pi-A}{2}\right)\cos\left(\frac{B-C}{2}\right) \qquad \because \quad A+B+C=\pi$$

$$\Rightarrow \quad 2\sin\frac{A}{2}\cos\frac{A}{2}=2\sin\frac{A}{2}\cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \quad \cos\frac{A}{2}=\cos\frac{B-C}{2} \text{ or } A = B - C \text{ ; But } A + B + C = \pi$$

Therefore $2B = \pi \Rightarrow B = \pi/2$ **Ans.(A)**

Illustration 15: If $A + B + C = \frac{3\pi}{2}$, then cos 2A + cos 2B + cos2C is equal to-

(A) 1 – 4cosA cosB cosC (B) 4 sinA sin B sinC

(C) 1 + 2cosA cosB cosC (D) 1 – 4 sinA sinB sinC

Solution : cos 2A + cos 2B + cos 2C = 2 cos (A + B) cos (A – B) + cos 2C

$$= 2\cos\left(\frac{3\pi}{2}-C\right)\cos(A-B)+\cos 2C \qquad \because A+B+C=\frac{3\pi}{2}$$

$= -2 \sin C \cos (A- B) + 1 - 2 \sin^2 C = 1 - 2 \sin C [\cos (A- B) + \sin C)$

$$= 1 - 2\sin C\left[\cos(A-B)+\sin\left(\frac{3\pi}{2}-(A+B)\right)\right]$$

$= 1 - 2 \sin C [\cos (A - B) - \cos (A + B)] = 1 - 4 \sin A \sin B \sin C$

Ans.(D)

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Do yourself - 14 :

1. If ABCD is a cyclic quadrilateral, then find the value of sinA + sinB – sinC – sinD

2. If $A + B + C = \frac{\pi}{2}$, then find the value of tanA tanB + tanBtanC + tanC tanA

If A + B + C = 180º, then prove that

3. $\sin2A + \sin2B - \sin2C = 4\cos A \cos B \sin C$

4. $\cos2A + \cos2B - \cos2C = 1 - 4\sin A \sin B \cos C$

5. $\sin A + \sin B - \sin C = 4 \sin\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

6. $\sin^2A + \sin^2B - \sin^2C = 2 \sin A \sin B \cos C$

7. $\cos^2A + \cos^2B + \cos^2C = 1 - 2 \cos A \cos B \cos C.$

8. $\cos^2A + \cos^2B - \cos^2C = 1 - 2 \sin A \sin B \cos C$

9. $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2}=1-2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

10. $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=1-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

11. $\sin(B + 2C) + \sin(C + 2A) + \sin(A + 2B) = 4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}.$

12. $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2}-1=4\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\sin\frac{\pi-C}{4}$

13. $\frac{\sin2A+\sin2B+\sin2C}{\sin A+\sin B+\sin C}=8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

14. $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4\sin A \sin B \sin C$

15. If x + y + z = xyz prove that

(a) $\frac{3x-x^3}{1-3x^2}+\frac{3y-y^3}{1-3y^2}+\frac{3z-z^3}{1-3z^2}=\frac{3x-x^3}{1-3x^2}\cdot\frac{3y-y^3}{1-3y^2}\cdot\frac{3z-z^3}{1-3z^2}$

(b) $x(1 - y^2)(1 - z^2) + y(1 - z^2)(1 - x^2) + z(1 - x^2)(1 - y^2) = 4xyz$

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18. MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC EXPRESSIONS :

(i) $a\cos\theta + b\sin\theta$ will always lie in the interval $[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$ i.e. the maximum and minimum values are $\sqrt{a^2+b^2}, -\sqrt{a^2+b^2}$ respectively.

(ii) Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta = 2ab$ where $a, b > 0$

(iii) $-\sqrt{a^2+b^2+2ab\cos(\alpha-\beta)} \le a\cos(\alpha+\theta) + b\cos(\beta+\theta) \le \sqrt{a^2+b^2+2ab\cos(\alpha-\beta)}$ where α and β areknown angles.

(iv) In case a quadratic in $\sin\theta$ & $\cos\theta$ is given then the maximum or minimum values can be obtained by making perfect square.

Illustration 16: Prove that : $-4 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right)+3 \le 10$, for all values of θ.

Solution : We have, $5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) = 5\cos\theta + 3\cos\theta\cos\frac{\pi}{3} - 3\sin\theta\sin\frac{\pi}{3} = \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta$

Since, $-\sqrt{\left(\frac{13}{2}\right)^2+\left(-\frac{3\sqrt{3}}{2}\right)^2} \le \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \le \sqrt{\left(\frac{13}{2}\right)^2+\left(-\frac{3\sqrt{3}}{2}\right)^2}$

$\Rightarrow \quad -7 \le \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \le 7$

$\Rightarrow \quad -7 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right) \le 7$ for all θ.

$\Rightarrow \quad -7+3 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right)+3 \le 7+3$ for all θ.

$\Rightarrow \quad -4 \le 5\cos\theta + 3\cos\left(\theta+\frac{\pi}{3}\right)+3 \le 10$ for all θ.

Illustration 17: Find the maximum value of $1 + \sin\left(\frac{\pi}{4}+\theta\right) + 2\cos\left(\frac{\pi}{4}-\theta\right)$ -

(A) 1 (B) 2 (C) 3 (D) 4

Solution : We have $1 + \sin\left(\frac{\pi}{4}+\theta\right) + 2\cos\left(\frac{\pi}{4}-\theta\right)$

$= 1 + \frac{1}{\sqrt{2}}(\cos\theta + \sin\theta) + \sqrt{2}(\cos\theta + \sin\theta) = 1 + \left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)(\cos\theta + \sin\theta)$

$= 1 + \left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)\cdot\sqrt{2}\cos\left(\theta-\frac{\pi}{4}\right)$

$\therefore$ maximum value $= 1+\left(\frac{1}{\sqrt{2}}+\sqrt{2}\right)\cdot\sqrt{2} = 4$ **Ans. (D)**

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Do yourself - 15 :

1. Find maximum and minimum value of $5\cos\theta + 3\sin\left(\theta + \frac{\pi}{6}\right)$ for all real values of θ.

2. Find the minimum value of $\cos\theta + \cos2\theta$ for all real values of θ.

3. Find maximum and minimum value of $\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2$.

4. Find the maximum and minimum values of

(i) $\cos2x + \cos^2x$ (ii) $\cos^2\left(\frac{\pi}{4} + x\right) + (\sin x - \cos x)^2$

5. If $\alpha + \beta = 90^\circ$, then find the maximum value of $\sin\alpha.\sin\beta$.

6. Find the maximum and minimum value of $1 + 2\sin x + 3\cos^2x$

7. Find the minimum value of $4\sec^2x + 9\operatorname{cosec}^2x$

8. Find the maximum and minimum value of $9\cos^2x + 48\sin x.\cos x - 5\sin^2x - 2$

9. Find the maximum and minimum value of $2\sin\left(\theta + \frac{\pi}{6}\right) + \sqrt{3}\cos\left(\theta - \frac{\pi}{6}\right)$

10. Find minimum value of (i) $3\sin^2x + 27\operatorname{cosec}^2x$ (ii) $27\sin^2x + 3\operatorname{cosec}^2x$

19. IMPORTANT RESULTS :

(i) $\sin\theta\sin(60^\circ - \theta)\sin(60^\circ + \theta) = \frac{1}{4}\sin3\theta$

(ii) $\cos\theta.\cos(60^\circ - \theta)\cos(60^\circ + \theta) = \frac{1}{4}\cos3\theta$

(iii) $\tan\theta\tan(60^\circ - \theta)\tan(60^\circ + \theta) = \tan3\theta$

(iv) $\cot\theta\cot(60^\circ - \theta)\cot(60^\circ + \theta) = \cot3\theta$

(v) (a) $\sin^2\theta + \sin^2(60^\circ + \theta) + \sin^2(60^\circ - \theta) = \frac{3}{2}$

(b) $\cos^2\theta + \cos^2(60^\circ + \theta) + \cos^2(60^\circ - \theta) = \frac{3}{2}$

(c) $\tan\theta + \tan(60^\circ + \theta) + \tan(120^\circ + \theta) = 3\tan3\theta$

(vi) (a) If $\tan A + \tan B + \tan C = \tan A\tan B\tan C$, then $A + B + C = n\pi$, $n \in I$

(b) If $\tan A\tan B + \tan B\tan C + \tan C\tan A = 1$, then $A + B + C = (2n + 1)\frac{\pi}{2}, n \in I$

(vii) $\cos\theta\cos2\theta\cos4\theta \ldots \cos(2^{n-1}\theta) = \frac{\sin(2^n\theta)}{2^n\sin\theta}$

(viii) (a) $\cot A - \tan A = 2\cot2A$ (b) $\cot A + \tan A = 2\operatorname{cosec}2A$

(ix) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots \sin(\alpha + \overline{n-1}\,\beta) = \frac{\sin\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

(x) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots + \cos(\alpha + \overline{n-1}\beta) = \frac{\cos\left\{\alpha + \left(\frac{n-1}{2}\right)\beta\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

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Do yourself - 16 :

1 Evaluate $\sin\frac{\pi}{n}+\sin\frac{3\pi}{n}+\sin\frac{5\pi}{n}+\ldots\ldots$ to n terms

2. Prove that : $\sin\theta + \sin3\theta + \sin5\theta +\ldots+ \sin(2n-1)\theta = \frac{\sin^2 n\theta}{\sin\theta}$

3. Find the average of $\sin2^\circ, \sin4^\circ, \sin6^\circ, \ldots, \sin180^\circ$

4. Prove that : $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}=\frac{1}{2}$

5. Find sum of the following series :

(a) $\cos\frac{\pi}{2n+1}+\cos\frac{3\pi}{2n+1}+\cos\frac{5\pi}{2n+1}+\ldots$ up to n terms.

(b) $\sin2\alpha + \sin3\alpha + \sin4\alpha +\ldots+ \sin n\alpha$, where $(n+2)\alpha = 2\pi$

6. If $S=\cos^2\frac{\pi}{n}+\cos^2\frac{2\pi}{n}+\ldots+\cos^2\frac{(n-1)\pi}{n}$, then S equals (where $n \geq 2$, $n \in N$)

(A) $\frac{n}{2(n+1)}$ (B) $\frac{1}{2(n-1)}$ (C) $\frac{n-2}{2}$ (D) $\frac{n}{2}$

Miscellaneous Illustration :

Illustration 18 : Prove that

$\tan\alpha + 2\tan2\alpha + 2^2\tan2^2\alpha + \ldots\ldots + 2^{n-1}\tan 2^{n-1}\alpha + 2^n \cot 2^n\alpha = \cot\alpha$

Solution : We know $\tan\theta = \cot\theta - 2\cot2\theta$ .....(i)

Putting $\theta = \alpha, 2\alpha, 2^2\alpha, \ldots\ldots\ldots\ldots$in (i), we get

$\tan\alpha = (\cot\alpha - 2\cot2\alpha)$

$2(\tan2\alpha) = 2(\cot2\alpha - 2\cot2^2\alpha)$

$2^2(\tan2^2\alpha) = 2^2(\cot2^2\alpha - 2\cot2^3\alpha)$

.....

$2^{n-1}(\tan2^{n-1}\alpha) = 2^{n-1}(\cot2^{n-1}\alpha - 2\cot2^n\alpha)$

Adding,

$\tan\alpha + 2\tan2\alpha + 2^2\tan^2\alpha + \ldots\ldots + 2^{n-1}\tan 2^{n-1}\alpha = \cot\alpha - 2^n\cot 2^n\alpha$

$\therefore$ $\tan\alpha + 2\tan2\alpha + 2^2\tan^2\alpha + \ldots\ldots + 2^{n-1}\tan 2^{n-1}\alpha + 2^n\cot 2^n\alpha = \cot\alpha$

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Illustration 19: If A,B,C and D are angles of a quadrilateral and $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\sin\frac{D}{2}=\frac{1}{4}$, prove that A = B = C = D = π/2.

Solution : $\left(2\sin\frac{A}{2}\sin\frac{B}{2}\right)\left(2\sin\frac{C}{2}\sin\frac{D}{2}\right)=1$

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right\}\left\{\cos\left(\frac{C-D}{2}\right)-\cos\left(\frac{C+D}{2}\right)\right\}=1$$

Since, A + B = 2π – (C + D), the above equation becomes,

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\right\}\left\{\cos\left(\frac{C-D}{2}\right)+\cos\left(\frac{A+B}{2}\right)\right\}=1$$

$$\Rightarrow \cos^2\left(\frac{A+B}{2}\right)-\cos\left(\frac{A+B}{2}\right)\left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{C-D}{2}\right)\right\}+1-\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{C-D}{2}\right)=0$$

This is a quadratic equation in $\cos\left(\frac{A+B}{2}\right)$ which has real roots.

$$\Rightarrow \quad \left\{\cos\left(\frac{A-B}{2}\right)-\cos\left(\frac{C-D}{2}\right)\right\}^2-4\left\{1-\cos\left(\frac{A-B}{2}\right).\cos\left(\frac{C-D}{2}\right)\right\}\geq 0$$

$$\left(\cos\frac{A-B}{2}+\cos\frac{C-D}{2}\right)^2\geq 4$$

$$\Rightarrow \quad \cos\frac{A-B}{2}+\cos\frac{C-D}{2}\geq 2\text{, Now both } \cos\frac{A-B}{2} \text{ and } \cos\frac{C-D}{2}\leq 1$$

$$\Rightarrow \quad \cos\frac{A-B}{2}=1 \,\&\, \cos\frac{C-D}{2}=1$$

$$\Rightarrow \quad \frac{A-B}{2}=0=\frac{C-D}{2}$$

$$\Rightarrow \quad A = B,\ C = D.$$

Similarly A = C, B = D ⇒ A = B = C = D = π/2

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